GGG–A–825a

April 3, 1967

SUPERSEDING
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June 23, 1964

FEDERAL SPECIFICATION

AUGER MACHINE, PIPE AND SEWER, POWER OPERATED

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers electric power operated auger machines used in buildings for removing stoppage and opening clogged pipe and sewer lines.

1.1.1 Federal specification coverage. Federal specifications do not include all varieties of the commodity as indicated by the title of the specification, or which are commercially available, but are intended to cover only those generally used by the Federal Government.

1.2 Classification.

1.2.1 Type. The auger machine covered by this specification shall be of one type as hereinafter specified.

2. APPLICABLE DOCUMENTS

2.1 Specifications and standards. The following specifications and standards, of the issue in effect on date of invitation for bids, or request for proposal form a part of this specification to the extent specified herein.

Federal Specifications:

J–C–175 —Cable Assembly, Power, Electrical, (3-Wire, 3-Prong, Grounding Plug Connector, for 125-Volt Equipment.
W–C–596 —Connector, Plug, Electrical; Connector, Receptacle, Electrical.
CC–M–636 —Motor, Alternating-Current, (Fractional Horsepower).
QQ–W–428 —Wire, Steel, High Carbon, Round, Uncoated for Mechanical Springs, (General Purpose).
QQ–W–470 —Wire, Steel, Carbon, Spring, Music.
UU–T–81 —Tags, Shipping and Stock.
PPP–B–601—Boxes, Wood, Cleated-Plywood.
PPP–B–621—Boxes, Wood, Nailed and Lock-Corner.
PPP–B–636—Box, Fiberboard.
PPP–B–640—Boxes, Fiberboard, Corrugated, Triple-Wall.
PPP–T–60 —Tape; Pressure-Sensitive Adhesive, Waterproof—for Packaging and Sealing.

Federal Standard:

Fed. Std. No. 123—Marking for Domestic Shipment (Civilian Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D.C., Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, and Seattle, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

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Military Specifications:

MIL-P-116 —Preservation, Methods of.
MIL-B-121 —Barrier Material, Greaseproofed, Waterproofed, Flexible.
MIL-L-10547—Liners, Case, and Sheet, Overwrap; Water-Vaporproof or Waterproof, Flexible.

Military Standards:

MIL-STD-105 —Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-109 —Quality Assurance Terms and Definitions.
MIL-STD-129 —Marking for Shipment and Storage.
MIL-STD-130 —Identification Marking of US Military Property.
MIL-STD-1186—Cushioning, Anchoring, Bracing, Blocking, and Waterproofing; With Appropriate Test Methods.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated the issue in effect on date of invitation for bids or request for proposal shall apply.

Governmental:

National Bureau of Standards Handbook:

H28—Screw-Thread Standards for Federal Services.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D.C., 20402).

3. REQUIREMENTS

3.1 Preproduction sample. When specified (see 6.2), before production is commenced, a sample of the finished item shall be submitted or made available to the contracting officer or his authorized representative for approval in accordance with 4.2. The approval of the preproduction sample authorizes the commencement of production, but does not relieve the supplier of responsibility for compliance with all applicable provisions of this specification. The preproduction sample shall be manufactured in the same facilities to be used for the manufacture of the production item.

3.2 Design. The auger machine shall include all the features, components, and accessories necessary for opening clogged sewer and drain lines. The auger machine shall be new, of the manufacturer's latest design, and shall be capable of meeting the performance requirements specified herein. The auger machine shall include a cable cage with flexible coil spring cable, electric motor and power supply cord, all mounted on a sturdy hand truck. The equipment shall be convenient for one man to operate. All parts subject to wear, breakage, or distortion shall be accessible for replacement, adjustment and repair.

3.2.1 Safety devices. Each machine shall be furnished with suitable safety devices of the latest type. Pulleys and other parts which are hazardous to the operator shall be suitably guarded. Ample protection against electrical shock shall be provided. When special safety devices are required, they shall be as specified (see 6.2).

3.2.2 Lubrication. When necessary to reduce wear and maintain efficiency, means shall be provided for lubrication except for permanently sealed-for-life mechanisms.

3.2.2 Interchangeability. All parts shall be manufactured to such standards as will permit replacement or adjustment without modification of parts or machine.

3.2.4 Material. Material shall be as specified herein. Material not definitely specified shall be of the quality necessary to produce a machine to meet the requirements specified herein.

3.3 Construction. Construction shall be free from any characteristics or defects that may render the machine unsuitable or inefficient for the purpose intended. The machine and its equipment shall be complete and ready for use in its intended application.

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3.3.1 Castings and forgings. All castings and forgings shall be free from defects, scale, and mismatching. Strength and other physical and chemical properties of the castings and forgings shall be adequate to meet the performance requirements specified herein. Castings and forgings shall not be subjected to any reclaiming process without the authorization of the procuring activity.

3.3.2 Welding, brazing and soldering. Welding, brazing and soldering shall be performed in accordance with the best commercial practice. In no event shall such processes be used for reclaiming any parts of the machine without authorization of the procuring agency.

3.3.3 Fastening devices. All screws, pins, bolts, and similar parts shall be installed with an adequate means for preventing loss of proper tightness and adjustment. All such parts, when subject to removal or adjustment, shall not be swaged, peened, staked, or otherwise permanently deformed.

3.3.4 Threads. All threaded parts shall be in the inch system and shall conform with the applicable requirements of NBS Handbook H28.

3.3.5 Surfaces. All surfaces of all castings, forgings, molded parts, stampings and welded parts shall be cleaned and free from sand, dirt, fins, sprues, scale, flux and other harmful or extraneous materials. External surfaces shall be smooth and all edges shall be either rounded or beveled.

3.3.6 Finish. Each auger machine shall be finished with a durable enamel, lacquer, or plating. Unless otherwise specified (see 6.2), any color normally furnished by the manufacturer will be acceptable.

3.4 Performance. The auger machine, with proper attachments and accessories, shall be capable of removing stoppage and opening clogged drain pipe and sewer lines up to 4 inches in diameter, and retrieving debris from such lines. The auger machine shall be capable of operating coil spring cable up to ⅝ inch in diameter in sewer line 4 inches in diameter at a distance of 200 feet. At this distance, the cable shall revolve at a minimum of 100 revolutions per minute.

3.5 Components.

3.5.1 Hand truck. The hand truck shall be of a sturdy, durable metal construction, mounted on rubber wheels. The frame shall provide suitable mountings for all components of the machine. The hand truck shall be provided with a suitable handle for easy maneuvering by one man.

3.5.2 Cable cage. The cable cage shall house and guide the flexible steel coil spring cleaning cable and shall be of sturdy steel or aluminum frame or steel wire mesh. Open wire mesh type shall be provided with a cover to prevent centrifugal splash. Means shall be provided for cleaning, oiling and draining of the coil spring and cage. Steel cages shall be heavily plated or enameled to deter corrosion. The cage shall have a short section of cable permanently mounted inside and shall be easily replaceable by another cage to provide additional length of cable by that means, if desired. When specified (see 6.2), additional cable cages may be ordered with coil spring cable sizes and lengths as shown in table I.

TABLE I.

Outside diameter of coil spring	Length of coil spring
Inch (minimum)	Feet (minimum)
½	50, 75, 100
⅝	50, 75, 100
¾	50, 75, 100

3.5.3 Coil spring. Coil spring cable shall be made of wire conforming to QQ–W–428, or music wire conforming to QQ–W–470. Cable shall have sufficient strength to meet the performance requirement of 3.4. The cable shall conform to table I for the diameter and length of cable specified (see 6.2). When specified (see 6.2) additional cables may be ordered in accordance with table I.

3.5.4. Electrical equipment.

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3.5.4.1 Motor. The electric motor shall operate on alternating current of 115 volts, 60 cycles, having a minimum rating of ⅓ horsepower, continuous duty at full rated load. Motor shall be either dripproof, splashproof, totally enclosed, or shall be mounted in an enclosure to prevent exposure to dripping or splashing water. Windings shall be impregnated to resist moisture. Bearings shall be sealed ball bearings, either permanently lubricated, or have accessible means for lubrication. Motor shall be wired for reverse operation. The motor shall conform to the applicable requirements of CC–M–636.

3.5.4.2 Motor control. The machine shall be equipped with a reverse switch, located convenient to the operator. A water resistant foot control "on-off" switch shall be provided.

3.5.4.3 Power supply cord. Unless otherwise specified (see 6.2), each machine shall be furnished with a 3-conductor flexible electric cord in accordance with type I of J–C–175, except for length. Length shall be a minimum of 6 feet, or as specified (see 6.2). The cord shall terminate in an appropriate 3-pole grounding plug conforming to style D of W–C–596. A mating two prong adapter with a protruding ground lead shall be furnished.

3.6 Cutter. The auger machine shall be equipped with a minimum of two cutter heads, for 3 and 4 inch drain or sewer line. The material shall be of a commercial grade suitable for the purpose intended. Means shall be provided so that the head may be replaced easily, and will not come loose during operation.

3.7 Couplers and adapters. Couplers and adapters, essential to the operation of the equipment, shall be furnished.

3.8 Optional equipment. Each auger machine shall be furnished with the following equipment, as specified, (see 6.2). Optional equipment furnished shall be adaptable and suitable to the auger machine furnished.

(a) Automatic feed (see 3.9.1).
(b) Flexible leader (see 3.9.2).
(c) Tool box (see 3.9.3).
(d) Additional cable heads (see 3.9.4).

3.8.1 Automatic feed. When specified (see 6.2), the auger machine shall be provided with an automatic feed mechanism which will allow the cleaning cable to automatically move into and out of a pipe or sewer line. Means shall be provided for disengaging the feed for manual operation.

3.8.2 Flexible leader. When specified (see 6.2), a flexible leader which may be attached to the machine cleaning cable shall be provided for clearing stoppage in "P" traps.

3.8.3 Tool box. When specified (see 6.2), the auger machine shall be provided with a metal tool box of sufficient size to hold cutters, tools, adapter and other accessories which may be furnished with the machine.

3.8.4 Additional cable heads. When specified (see 6.2), various types of heads shall be provided for doing specific sewer or drain cleaning operations.

3.9 Identification marking.

3.9.1 Civil agencies. The auger machine shall be provided with the manufacturer's nameplate or marked in a plain and permanent manner so that the source of manufacture may be readily determined.

3.9.2 Military activities. The auger machine shall be marked in accordance with MIL–STD–130.

3.10 Handbook of instructions and parts list. One copy of parts list, and handbook of instructions for operation, care and maintenance of the auger machine shall be furnished.

3.11 Workmanship. Workmanship shall be of the quality necessary to produce a machine and equipment of good appearance, having stability and efficient operating characteristics found in first-class commercial items of a similar nature.

4. QUALITY ASSURANCE PROVISIONS

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4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Preproduction sample inspection. When a preproduction sample is required (see 3.1), it shall be examined for defects listed in 4.3.2.1; dimensions specified and tested as specified in 4.3.2.2; and shall be tested as specified in operational tests 4.4.1 and functional tests 4.4.2. Failure to meet requirements of specified tests shall be cause for rejection.

4.3 Inspection for acceptance. Inspection shall be made in accordance with the provisions set forth in MIL–STD–105, except where otherwise indicated hereinafter. The inspection terms and definitions shall be interpreted in accordance with MIL–STD–109.

4.3.1 Inspection of components and material. In accordance with 4.1 above, the supplier is responsible for insuring that components and materials are manufactured, tested, and inspected in accordance with the requirements of referenced subsidiary specifications and standards, to the extent specified or, if none, in accordance with this specification.

4.3.2 End item inspection. The inspection lot shall consist of all completed auger machines offered for inspection at one time. The sample unit shall be one complete auger machine.

4.3.2.1 Visual examination. Examination of the auger machine shall be in accordance with the classification of defects set forth in table II. The inspection level shall be level II with an acceptable quality level (AQL) of 2.5 for major and 6.5 for total defects, expressed in terms of defects per hundred units.

TABLE II. Classification of defects

Examine	Defect	Classification	
		Major	Minor
Finish	Type of finish not as specified or blistered, peeled, chipped or area of no thin film		X
	Not finished where required	X	
	Component or assembly that is painted or otherwise coated where serviceability may be affected.		X
Materials	Component not fabricated of the specified material or the type material authorized in the approved sample	X	
Design	Any characteristic not in accordance with the specified requirements	X	
Construction and workmanship, general (applicable to all components and assemblies)	Part missing or not specified type and size	X	
	Fractured, split, punctured, dented, deteriorated, bowed, malformed, or otherwise impaired affecting serviceability.	X	
	Defective or otherwise damaged, but not affecting serviceability		X
	Misplaced, loose or not in proper alignment affecting serviceability	X	
	Sharp burr, sliver or splinter that may be injurious to personnel or may affect serviceability	X	
	Functioning component that requires abnormal force to operate		X

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TABLE II. Classification of defects—(Cont'd)

Examine	Defect	Classification	
		Major	Minor
	Not connected or joined as specified, affecting serviceability	X	
	Any adjustable assembly that cannot be adjusted or is not properly adjusted to perform the function intended	X	
	Any component not readily accessible for servicing where required		X
Welding and brazing (where required)	Not type specified, missing, incomplete, burn holes, cracked, fractured or otherwise not fused	X	
	Slag inclusion, slight undercut, not smooth and uniform, scale or flux deposits not removed.	X	
Soldering (when applicable)	Missing, not adherent or incomplete	X	
	Not clean (flux or flux residue not removed), not smooth (surface not finished neatly) or pin holes in solder		X
Bolts, nuts, screws, studs, and other types of threaded fasteners	Missing, not type specified, broken, stripped, fractured or loose, not affecting serviceability		X
Electrical assembly	Wiring cut, abraded, not properly joined, loose at terminal or not inclosed in specified conduit where required	X	
	Adequate slack not provided for wiring to relieve strain or excessive insulation stripped from wiring		X
Lubrication (where applicable)	Not lubricated where required	X	
	Insufficiently lubricated		X
	Not specified type lubrication fitting, affecting serviceability	X	
	Lubrication fitting not readily accessible for servicing		X
Assembly	Component not properly assembled or secured, but not affecting serviceability e.g., lockwasher missing from metal fastener		X
	Unit perceptibly out of square or alignment	X	
Marking, instructions, manual	Missing, incomplete, not legible		X

4.3.2.2 Dimensional examination. An examination shall be made of cleaning cable and power supply cord for compliance with dimensions specified. Any deviation therefrom shall constitute a defect. The inspection level shall be S–2 with an AQL of 2.5 expressed in terms of defects per hundred units.

4.4 Tests.

4.4.1 Operational testing. Each auger machine furnished under this specification shall be connected to power and operational tested. In the course of operational testing, determination shall be made to establish compliance with all performance requirements applicable to the fully assembled item.

4.4.2 Functional testing. Auger machines submitted for testing shall be tested for compliance to 3.4. A cleaning cable 200 feet in length with a 4 inch cutter head shall be used. The cleaning cable shall be of the same material and type furnished with the machine. After inserting in a 4 inch sewer line, determine that the cable and the cage rotate properly without buckling the cable in the cage or of having any other malfunction.

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4.5 Inspection of preparation for delivery requirements. An inspection shall be made to determine that packaging, packing and marking requirements are in compliance with section 5 of this specification. Defects shall be scored in accordance with table III. For examination of contents, the sample unit shall be one shipping container fully prepared for delivery selected just prior to the closing operations. Defects of closure listed shall be examined on shipping containers fully prepared for delivery. The lot size shall be the number of shipping containers in the end item inspection lot. The inspection level shall be S–2 and the AQL shall be 4.0 defects per hundred units.

TABLE III. Classification of preparation for delivery defects.

Examine	Defect
Markings (exterior and interior)	Omitted; incorrect; illegible; or improper size, location, sequence, or method of application.
Materials	Any component missing or damaged.
Workmanship	Inadequate application of components such as incomplete closure of container flaps, loose strapping, inadequate stapling.
	Bulging or distortion of container
Preservation	Preservatives not applied.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A, B, or C, as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Disassembly. Disassembly shall be limited to the desired reduction in cube and to parts vulnerable to damage or loss. Removed parts shall be matchmarked to facilitate reassembly. Matchmarking information shall be printed on tags conforming to UU–T–81, type A.

5.1.1.2 Preservation application. Cutters, adapters, and exposed bare metal surfaces shall be coated with type P–1 preservative of MIL–P–116.

5.1.1.3 Unit packaging. Auger machine shall be preserved and packaged in accordance with method I of MIL–P–116. External openings of the electric motor shall be covered with material conforming to MIL–B–121, grade A, class 2; secured with tape conforming to PPP–T–60, class 1. Tension on all drive belts shall be released. Disassembled parts and extra cable shall be packaged in a close-fitting box conforming to PPP–B–636, class domestic, type CF, variety SW, grade 275. The box shall be closed in accordance with the appendix to the box specification.

5.1.2 Level B, civil agencies.

5.1.2.1 Disassembly. Disassembly shall be limited to the desired reduction in cube and to parts vulnerable to damage or loss. Removed parts shall be matchmarked to facilitate reassembly. Matchmarking information shall be printed on tags conforming to UU–T–81, type A.

5.1.2.2 Unit packaging. The electric cable shall be compactly wound on the bracket and secured. The tension on all drive belts shall be released. Disassembly parts and extra cable shall be packaged in a close-fitting box conforming to PPP–B–636, class domestic. The box shall be closed in accordance with the appendix to the box specification.

5.1.3 Level C. The augers shall be preserved and packaged to afford adequate protection against corrosion, deterioration, and damage during shipment from the supply source to the first receiving activity for immediate use. The suppliers commercial practice will be acceptable when it fulfills the above requirements.

5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.2).

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5.2.1 Level A. Each complete auger preserved and packaged as specified in 5.1.1 shall be packed in a close-fitting box conforming to PPP–B–621, class 2, style 2. A case liner conforming to MIL–L–10547 shall be provided. Internal cushioning, blocking, anchoring, and bracing shall be in accordance with MIL–STD–1186. The box shall be closed and strapped in accordance with the appendix to the box specification. Containers shall be uniform size and shape of minimum tare and cube.

5.2.2 Level B. Each complete auger shall be packed in a close-fitting box conforming to PPP–B–601, domestic type; PPP–B–621, class 2; PPP–B–636, class domestic, or PPP–B–640, class 2, grade A. Cushioning, blocking and bracing shall be provided. The box shall be closed in accordance with the appendix to the box specification.

5.2.3 Level C. The augers shall be packed to insure carrier acceptance and safe delivery at destination at the lowest rates, in containers complying with the rules and regulations applicable to the mode of transportation.

5.3 Marking.

5.3.1 Civil agencies. In addition to markings required by the contract or order, the interior packages and shipping containers shall be marked in accordance with Fed. Std. No. 123.

5.3.2 Military activities. In addition to markings required by the contract or order, the interior packages and shipping containers shall be marked in accordance with MIL–STD–129.

6. NOTES

6.1 Intended use. The auger machine covered by this specification is intended for use in buildings for opening clogged sewer and pipe lines and for periodic maintenance cleaning of such lines.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Preproduction sample, if required (see 3.1).
(c) Special safety devices, if required (see 3.2.1.
(d) Color of machines, if different (see 3.3.6).
(e) Additional cable cages, if required (see 3.5.2).
(f) Size and number of coil springs required (see 3.5.3).
(g) Length of power supply cord, if different (see 3.5.4.3).
(h) Optional equipment, if required (see 3.9).
(i) Selection of applicable level of preservation, packaging and packing required (see 5.1 and 5.2).
(j) Special marking, if required (see 5.3).

MILITARY INTEREST:

Custodians:

Army—WC
Navy—YD
Air Force—82

Review activities:

Army—WC
Navy—YD
Air Force—82

Civil Agency Interest

GSA—FSS

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